

Master Internship Proposal: Theoretical Foundations and Deep Learning Algorithms for Equivariant Tomographic Imaging

Supervision:

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Location: CREATIS Research Lab, Lyon, France

Start time: Flexible, January to May 2027.

Context and Motivation

In fields like cryo-electron tomography (cryo-ET), high-resolution 3D volume reconstruction is fundamentally limited by incomplete data acquisition. Because the sample cannot be rotated a full 360 degrees, the resulting measurements suffer from missing frequencies, which can be modeled mathematically as an angular binary Fourier mask. While modern deep learning techniques have shown immense promise in filling these missing data regions, they frequently lack rigorous mathematical guarantees for exact recovery. This internship addresses this critical gap by exploring the theoretical properties of equivariant imaging [1]. The ambition is to develop an unsupervised deep learning framework where structural measurement consistency and the compatibility of synthesized rotations inherently guarantee exact reconstruction.

Internship Objectives

The intern will drive both the mathematical formulation and the algorithmic implementation of a novel tomographic reconstruction method. Core responsibilities include:

- **Theoretical Analysis:** Establish sufficient mathematical conditions for exact volume reconstruction. This involves analyzing the commutation of Fourier masks and the injectivity of operators on admissible signal sets.
- **Distribution Matching:** Demonstrate theoretically and empirically how matching the population distribution of generated measurements to the true measurement distribution enforces structural compatibility.
- **Algorithm Development:** Design and implement a finite-data training algorithm that leverages independent Haar-uniform rotations to generate synthetic measurements.
- **Unsupervised Optimization:** Formulate a training objective combining empirical measurement-consistency loss with a distribution discrepancy metric, eliminating the need for paired ground-truth volumes.

Practical Application: Advancing Cryo-ET

Bridging theory and practice is a cornerstone of this role. In cryo-ET, the "missing wedge" artifacts severely distort the recovered biological structures. The framework developed during this internship will directly tackle this physical limitation. By rotating the reconstructed volume and subsequently applying the fixed acquisition mask, the algorithm will generate synthetic measurements that constrain the

unmeasured frequencies. The intern will be responsible for pushing the continuous, noiseless theoretical models toward practical deployment, investigating how grid discretization, finite batch sizes, and physical noise impact the stability and accuracy of the exact-recovery guarantees.

Candidate Profile

We are seeking a curious, highly motivated Master’s student looking to engage with real-world scientific problems.

- **Background:** A strong foundation in linear algebra, functional analysis, applied mathematics, machine learning, or signal processing is required.
- **Technical Skills:** Experience with programming and scientific computing is essential. Knowledge of modern deep learning frameworks like PyTorch, alongside familiarity with GPU computing, will be highly beneficial.
- **Mindset:** An interest in interdisciplinary research and connecting theoretical modeling with practical experiments is crucial.

Why Join Us?

This PhD offers the opportunity to work on fundamental questions with real-world impact [2], to connect theory with experimental science, and to grow in a supportive and ambitious research environment. It provides a strong foundation for pursuing a career in academic or industrial research.

How to apply? Please send a curriculum and your academic records to

valentin.debarnot@creatis.insa-lyon.fr

This position will remain open until a suitable candidate is found.

References

- [1] Dongdong Chen, Julián Tachella, and Mike E Davies. Equivariant imaging: Learning beyond the range space. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 4359–4368. IEEE, 2021.
- [2] Vinith Kishore, Valentin Debarnot, Ricardo D Righetto, Benjamin D Engel, and Ivan Dokmanić. Icecream: High-fidelity equivariant cryo-electron tomography. *Acta Crystallographica Section D*, 2026.